Matrix - AutomationLogicAndWebhookTriage - Engine

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Document ID:     Matrix - AutomationLogicAndWebhookTriage - Engine
Naming Format:   [CATEGORY] - [PURPOSE] - [TIER]
Category:        Matrix (Algorithmic Evaluation & Incident Governance)
Purpose:         AutomationLogicAndWebhookTriage
Tier / Level:    Engine (Event Priority & Fail-Safe Architecture)
Status:          Locked 🔒 (Master Baseline)
Target Audience: Lead TechOps Specialists, Automation Engineers, Systems Integrators
Ecosystem Scope: Shopify Flow, Shopify Webhooks API, AWS SQS / Redis DLQ, 3PL WMS (ShipBob / ShipBob WMS Gorgias
Business Unit:   Synapse Botanicals (Automation & Systems Infrastructure)
```

1. Executive Summary & Event-Driven Governance Philosophy

1.1 Purpose & Architectural Scope

In distributed modern commerce, event-driven architectures execute mission-critical operations: real-time fraud evaluation, dynamic multi-node fulfillment routing, inventory synchronization, and downstream ERP/CRM record propagation. When an automation pipeline fails or executes out of sequence, the resulting failures manifest as duplicate warehouse shipments, corrupted inventory ledgers, dropped customer orders, and customer service escalation cascades.

This document establishes the authoritative technical decision engine governing **Shopify Flow execution priority**, **HTTP webhook response code triage**, **automated retry curves**, **circuit-breaker thresholds**, and **recursion prevention guardrails** for Synapse Botanicals.

1.2 The Three Core Triage Axioms

1. **Zero Silent Data Loss:** Every inbound and outbound webhook event must either resolve with an HTTP `200/202` acknowledgment or log a persistent failure payload in an observable Dead-Letter Queue (DLQ).
2. **Deterministic State Isolation:** An automated action must evaluate explicit, mutually exclusive preconditions before modifying order, customer, or inventory states.
3. **Idempotent Ingestion & Replay:** Every webhook consumer and automation handler must track unique event identifiers (`X-Shopify-Webhook-Id` or GraphQL GIDs) to guarantee that duplicate deliveries produce zero side effects.

> **NOTE**
> **System Architecture Dependency:**
> This decision engine operationalizes the automation standards specified in **SOP - BusinessLogicAutomationAndWebhooks - Parent (Domain 4.1)** – **Section 3.1** and directly enforces the fulfillment lock gates modeled in **Matrix - OrderStatePointOfNoReturn - Master (Domain 1.2)** – **Section 2.0**.

2. Shopify Flow Priority Resolution & Race Condition Governance

When an order event occurs, multiple Shopify Flow workflows trigger concurrently. Without deterministic priority arbitration, asynchronous race conditions occur—such as routing an order to a 3PL node before fraud analysis can apply an administrative hold.

2.1 The Four-Tier Execution Priority Ladder

Workflows must be designed, tagged, and ordered according to the four-tier priority hierarchy detailed below:

Priority Tier	Classification	Target Workflows	Maximum Execution Latency	Downstream Dependency
P0 (Critical)	Risk & Integrity	Fraud Analysis, Order Hold Injection, Sanctions/Address Scrubbing	≤ 1,500 ms	**Blocks all P1–P3 execution.** Prevents 3PL fulfillment release.
P1 (High)	Node Routing	Multi-Location Inventory Allocation (`LOC-PA-01` vs. `LOC-NV-02`), Heavy-Basket Packaging Routing	≤ 3,000 ms	Unlocks fulfillment order assignment and 3PL routing tags.

Priority Tier	Classification	Target Workflows	Maximum Execution Latency	Downstream Dependency
P2 (Operational)	Operations & CRM	B2B Wholesale Tagging, Pre-Order Metadata Stamping, Gorgias VIP Ticket Escalation	≤ 10,000 ms	Non-blocking. Informs warehouse pack slips and support tiers.
P3 (Background)	Marketing & Retention	Loyalty Tier Adjustments, Delayed Review Sequences (Wait 14 Days), Klaviyo Lifecycle Sync	Asynchronous / Delayed	Completely decoupled from order fulfillment lifecycle.

2.2 Event Trigger vs. Action Conflict Resolution Matrix

Trigger Event	Concurrently Executing Workflows	Potential Race Condition / Conflict	Structural Idempotency & Guard Pattern	Priority Resolution
`Order created`	• Flow 4.1: Fraud Risk Hold • Flow 4.2: Heavy Basket Tagging (`SYN-TMBL-MO`) • 3PL Ingestion Webhook	The 3PL webhook consumes the order and drops it to the warehouse floor before Flow 4.1 can inject `flow-hold:fraud-review` .	Add a mandatory 5-minute buffer delay in 3PL fulfillment release rules, OR require condition: `Order tags does NOT contain flow-hold:*` .	**P0 Wins:** Fraud hold applies immediately; 3PL webhook ignores held orders.
`Order updated`	• Flow 4.3: Address Verification • Customer Profile Auto-Sync • ERP Line-Item Reconciliation	Address edit in Shopify admin fires `Order updated` , triggering an auto-tag workflow that re-fires `Order updated` indefinitely.	Strict Tag Exclusion Guard: Precondition must verify `Order tags does NOT contain sys:sync-locked` . Apply tag on pass.	**Loop Terminated:** Single-pass execution guaranteed via namespace sentinel tag.
`Inventory level updated`	• Low Stock Warning (`< 50` units) • Auto-Draft Purchase Order Flow • Bundle Disassembly Automation	Simultaneous decrements across Allentown (`LOC-PA-01`) and Reno (`LOC-NV-02`) trigger multiple concurrent inventory threshold webhooks.	Evaluate `inventory_levels` by discrete Location ID rather than aggregated global variant quantity. Apply 60s throttle.	**Node-Level Isolation:** Inventory events isolate to specific fulfillment node.
`Customer created`	• Wholesale Verification • Welcome Series Sync • VIP Loyalty Classification	Customer tagged as `wholesale-pending` by Admin, simultaneously tagged as `retail-welcome` by Klaviyo integration.	Flow condition hierarchy: `If customer email matches B2B domain -> Add Tag 'cs:b2b-prospect' -> Exit Flow immediately` .	**Deterministic Exit:** B2B classification halts retail drip sequences.

> **NOTE**
>
> **System Architecture Dependency (Order Lifecycle):**
> For complete technical governance on how P0 Fraud Holds interact with warehouse fulfillment order lines and manual customer address mutations, consult **Matrix - OrderStatePointOfNoReturn - Master (Domain 1.2)** – **Section 2.0**.

3. HTTP Webhook Triage & Status Code Response Matrix

Shopify webhooks expect an HTTP `2xx` success response within **5 seconds (5,000 ms)**. If a receiving server returns a non-2xx status code or times out, Shopify's delivery pipeline marks the execution as failed and enters an automated retry cycle.

The matrix below defines the deterministic operational protocol for each HTTP status code encountered across the Synapse Botanicals webhook ingestion pipeline:

HTTP Status Code	Response Classification	Root Cause Pattern at Synapse Botanicals	Shopify Delivery Behavior	Automated Triage Action	Recovery SLA	Preventative Engineering Safeguard
`200 OK`	Nominal Success	Payload received, HMAC validated, and business logic processed synchronously.	Delivery marked complete. Delivery log closed.	Archive raw payload header in 30-day cold log buffer.	Instantaneous	Maintain handler latency ≤ 800 ms.

HTTP Status Code	Response Classification	Root Cause Pattern at Synapse Botanicals	Shopify Delivery Behavior	Automated Triage Action	Recovery SLA	Preventative Engineering Safeguard
`202 Accepted`	Asynchronous Success	Payload received, HMAC verified, and raw JSON spooled into message queue (SQS/PubSub/Redis) for background consumption.	Delivery marked complete. Handshake terminated safely.	Enqueue message to background worker pool; immediately release HTTP connection.	Instantaneous	Recommended architecture for all high-volume endpoints to defeat the 5s timeout.
`400 Bad Request`	Client Payload Error	Malformed JSON body or invalid HTTP headers generated by intermediate reverse proxy.	Shopify retries according to the 19-attempt backoff curve.	Inspect Cloudflare / edge proxy logs for body truncation or header stripping.	1 Hour	Bypass edge WAF inspection for trusted Shopify webhook IP ranges.
`401 Unauthorized`	Authentication Failure	Missing or corrupt API bearer token; failed HMAC-SHA256 signature verification.	Shopify retries according to the 19-attempt backoff curve.	**Trigger P1 PagerDuty Alert:** Rotate API keys in private app settings; re-sync shared secret in middleware.	15 Minutes	Automated credential rotation script with dual-secret overlapping validation windows.
`403 Forbidden`	Scope / Permission Loss	App access token lacks required Admin API scopes (e.g., `read_fulfillments` revoked).	Shopify retries according to the 19-attempt backoff curve.	Re-authorize app permissions via Shopify Admin Partner portal.	30 Minutes	CI/CD permission linting during app deployment pipeline.
`404 Not Found`	Broken Route / Endpoint	Ingestion URL path changed or deprecated (e.g., `/api/webhooks/orders` returns 404).	Shopify retries according to the 19-attempt backoff curve.	Roll back endpoint routing in deployment ingress; restore route alias in server config.	15 Minutes	Immutable API routing contracts; deprecation redirects (`307 Temporary Redirect` not supported by Shopify—must return 200).
`422 Unprocessable Entity`	Business Logic Rejection	Schema mismatch (e.g., Order includes unmapped SKU like legacy placeholder `SYN-STK` instead of `SYN-FSTK-MY`).	Shopify retries according to the 19-attempt backoff curve.	Middleware catches rejection, logs payload to DLQ, and generates Ops ticket with failed SKU GID.	4 Hours	Catalog validation gate on Shopify Admin preventing publish of unmapped SKUs.
`429 Too Many Requests`	Rate Limit Breached	Downstream receiving server or 3PL WMS API hit burst limit during a flash sale.	Shopify retries according to the 19-attempt backoff curve.	Shift webhook receiver to asynchronous queue ingestion (`202 Accepted` pattern).	30 Minutes	Implement token-bucket rate smoothing in edge ingestion layer.
`500 Internal Server Error`	Uncaught Server Exception	Receiving application crashed (unhandled null exception on missing shipping address field).	Shopify retries according to the 19-attempt backoff curve.	Ingestion layer captures raw stack trace; fallback worker captures event in Redis buffer.	1 Hour	Defensive payload parsing (`payload.shipping_address?.zip ?? ""` with optional chaining).

HTTP Status Code	Response Classification	Root Cause Pattern at Synapse Botanicals	Shopify Delivery Behavior	Automated Triage Action	Recovery SLA	Preventative Engineering Safeguard
`502 Bad Gateway`	Reverse Proxy Failure	Node.js process crashed or NGINX upstream server failed to respond.	Shopify retries according to the 19-attempt backoff curve.	PM2 / Kubernetes auto-heals pod; check container memory limits and OOM kills.	15 Minutes	Container memory headroom monitoring with alert threshold at 80% RAM utilization.
`503 Service Unavailable`	Service Outage	3PL WMS undergoing scheduled database maintenance or server failure.	Shopify retries according to the 19-attempt backoff curve.	Trip **Circuit Breaker** (Section 5) to buffer webhooks in intermediate queue.	2 Hours	Multi-region failover and decouple synchronous 3PL dependencies via staging tables.
`504 Gateway Timeout`	Execution Timeout	Ingestion handler blocked on synchronous database write; execution exceeded **5,000 ms**.	Shopify terminates connection and retries according to backoff curve.	Offload all database transactions and downstream API calls to background queues.	30 Minutes	Enforce hard 2,000 ms timeout on HTTP handler; force async handoff.

> **NOTE**
> **System Architecture Dependency (Webhook Security & Kill-Switch):**
> For complete instructions on implementing timing-safe HMAC SHA-256 verification and deploying the Emergency Webhook Kill-Switch, refer to **SOP - BusinessLogicAutomationAndWebhooks - Parent (Domain 4.1)** – **Section 4.0**.

4. The 48-Hour Webhook Retry Curve & The 19-Attempt Dropout Cliff

Shopify's webhook infrastructure implements an automated retry policy designed to bridge transient network disruptions. Understanding this schedule is vital: failure to resolve an outage before the retry schedule expires results in the **silent, permanent deletion** of the webhook subscription.

4.1 The 19-Attempt Exponential Backoff Schedule

Shopify attempts redelivery **19 times over an approximate 48-hour window**, utilizing an exponential backoff algorithm with randomized jitter.

```
Delivery Attempt Timeline (48-Hour Window)
[T+0m]──[T+5m]──[T+15m]──[T+30m]──[T+1h]──[T+2h]──[T+4h]──...──[T+48h] ──> [SUBSCRIPTION TERMINATED]
  |       |        |        |        |        |        |              |
Att 1   Att 2    Att 3    Att 4    Att 5    Att 6    Att 7          Att 19
```

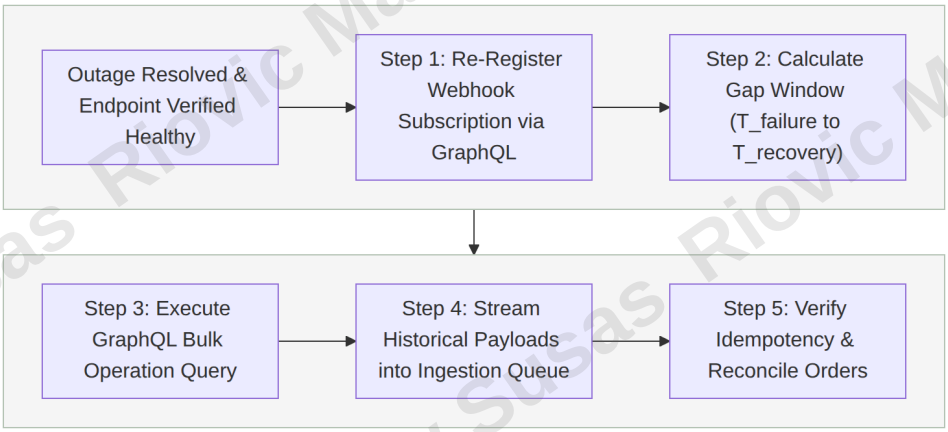
Attempt Number	Time Offset from Incident	Cumulative Delay	Interval Category	Escalation & Triage Protocol
1	0 min	0 min	Initial Trigger	Real-time failure. Subscribed endpoint returns non-2xx or times out (>5s).
2	+5 min	5 min	Immediate Retry	Automated retry. Logged as transient network blip.
3	+10 min	15 min	Short Backoff	Secondary retry. If failed, system raises warning log.
4	+15 min	30 min	Short Backoff	Monitoring bot detects 4 consecutive failures across single endpoint.
5	+30 min	1.0 hour	Intermediate Backoff	**Tier 1 Slack Alert:** `#ops-tech-alerts` notified of recurring failure.
6	+1.0 hour	2.0 hours	Intermediate Backoff	Automated endpoint health check (HTTP GET `/healthz`) initiated.
7	+1.5 hours	3.5 hours	Intermediate Backoff	Engineer on-call verifies receiving server status and database connectivity.
8	+2.0 hours	5.5 hours	Extended Backoff	Middleware shifts endpoint into canary mode.

Attempt Number	Time Offset from Incident	Cumulative Delay	Interval Category	Escalation & Triage Protocol
9	+2.5 hours	8.0 hours	Extended Backoff	Manual inspection of failed payload schema against current Shopify API version.
10	+3.0 hours	11.0 hours	Half-Day Warning	**Tier 2 PagerDuty Incident:** Endpoint down for >10 hours during active sales.
11	+3.5 hours	14.5 hours	Deep Backoff	Investigation of upstream DNS, Cloudflare SSL certs, or token expiration.
12	+4.0 hours	18.5 hours	Deep Backoff	Circuit Breaker automatically shifts traffic to Dead-Letter Queue.
13	+4.5 hours	23.0 hours	Day 1 Milestone	Outage extends past 24 hours. Formal incident bridge established.
14	+5.0 hours	28.0 hours	Critical Window	Order accumulation in retry buffer exceeds 500 records.
15	+5.5 hours	33.5 hours	Critical Window	3PL advised of delayed fulfillment order drop; manual fulfillment held.
16	+6.0 hours	39.5 hours	Final Warnings	Management notified of impending webhook subscription cancellation.
17	+4.0 hours	43.5 hours	Final Warnings	Staging backup server stood up if primary server recovery fails.
18	+3.0 hours	46.5 hours	Terminal Retry	Final automated attempts prior to Shopify subscription purge.
19	+1.5 hours	48.0 hours	**Point of No Return**	**FINAL ATTEMPT.** If failed, Shopify permanently deletes webhook registration.

4.2 The Silent Dropout Cliff & Post-Dropout Recovery

When Attempt 19 fails, Shopify automatically deletes the webhook subscription. Shopify sends an administrative email to the store owner's technical contact address, but **does not emit an API event or automated webhook alert**.

If an operations team does not actively monitor subscription health, the integration goes completely dark. Subsequent orders, edits, and cancellations occur in Shopify without any data transmitting to downstream systems.



Step 1: Re-Register Webhook Subscription via GraphQL Admin API

Execute the following mutation to re-establish the severed subscription immediately:

```
mutation reRegisterWebhookSubscription {
  webhookSubscriptionCreate(
    topic: ORDERS_CREATE
    webhookSubscription: {
      callbackUrl: "https://api.synapsebotanicals.com/v1/webhooks/orders/create"
      format: JSON
      includeFields: ["id", "name", "email", "line_items", "shipping_address", "tags", "total_price"]
    }
  ) {
    userErrors {
      field
      message
    }
```

```
    webhookSubscription {
      id
      topic
      endpoint {
        __typename
        ... on WebhookHttpEndpoint {
          callbackUrl
        }
      }
    }
  }
}
```

Step 2: Historical Order Backfill Query
To backfill every order dropped during the outage window without triggering duplicate fulfillment requests, execute a bulk query bounded by the exact downtime timestamps:

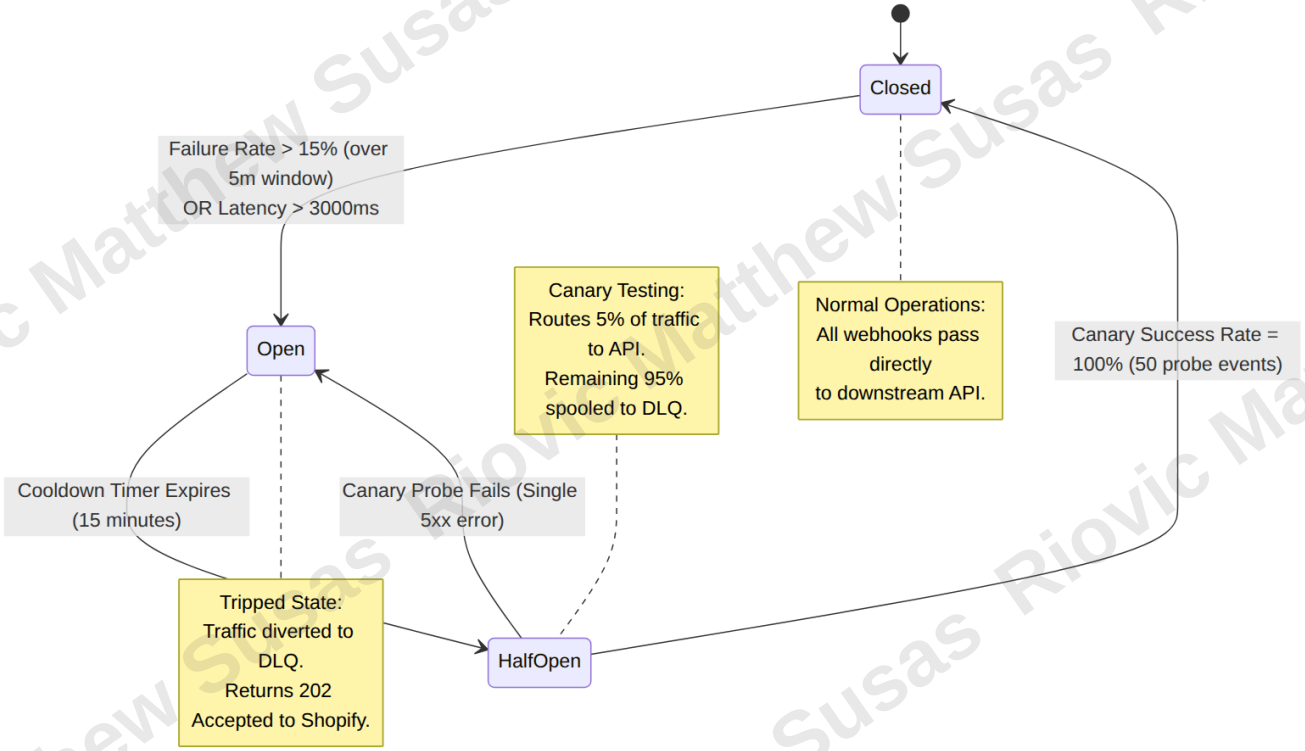
```
mutation backfillDroppedOrders {
  bulkOperationRunQuery(
    query: """
    {
      orders(query: "created_at:>=2026-10-15T08:00:00Z AND created_at:<=2026-10-17T08:00:00Z") {
        edges {
          node {
            id
            name
            createdAt
            displayFulfillmentStatus
            tags
            lineItems {
              edges {
                node {
                  sku
                  quantity
                  variant {
                    id
                    barcode
                  }
                }
              }
            }
          }
        }
      }
    }
    """
  ) {
    bulkOperation {
      id
      status
    }
    userErrors {
      field
      message
    }
  }
}
```

---

5. Circuit-Breaker State Machine & Ingestion Throttling Engine

To prevent cascading system collapse when downstream services (e.g., 3PL WMS, Gorgias CRM) degrade, Synapse Botanicals implements an automated **Three-State Circuit Breaker Pattern** at the webhook ingestion proxy.

5.1 Three-State Circuit Breaker Model



5.2 Threshold Configuration & Metric Gates

Parameter	Operational Threshold	Metric Definition	Action Triggered Upon Breach
Error Rate Trip Threshold	≥ 15.0%	Failed responses (`5xx` or timeouts) divided by total requests over a rolling 5-minute window.	**Trip Circuit to `OPEN`:** Cease forwarding to downstream endpoint. Return HTTP `202 Accepted` to Shopify and spool payloads directly to AWS SQS / Redis Dead-Letter Queue.
Consecutive Failure Threshold	10 consecutive requests	Ten sequential non-2xx responses or connection timeouts regardless of volume.	**Immediate Emergency Trip to `OPEN`:** Eliminates risk of connection pool exhaustion.
Latency Degradation Threshold	≥ 3,000 ms average	Rolling P95 latency across downstream endpoint over 3 consecutive minutes.	**Preemptive Throttling:** Downgrade synchronous processing; spool 50% of traffic to queue to alleviate downstream load.
Cooldown Duration	15 minutes	Elapsed duration the circuit remains in `OPEN` state before probing health.	**Transition to `HALF-OPEN`:** Releases canary probe batch.
Canary Evaluation Volume	50 requests (or 5% traffic)	Discrete batch of real incoming events routed to endpoint in `HALF-OPEN` state.	If all 50 succeed → Reset to `CLOSED`. If any 1 fails → Reset to `OPEN` for 30 minutes.
Maximum DLQ Retention	14 days	Time-to-live (TTL) for buffered payloads residing in Dead-Letter Queue.	Automated alert at 7-day mark if queue depth > 0.

6. Canonical Tag Namespace Schema & Recursion Guardrails

Infinite recursion loops in Shopify Flow occur when an automation responds to an `Order updated` trigger by appending a tag, which Shopify evaluates as a new update event, re-triggering the same workflow indefinitely. This saturates the shop's API rate limit, bloats order timelines with thousands of log entries, and exhausts server memory.

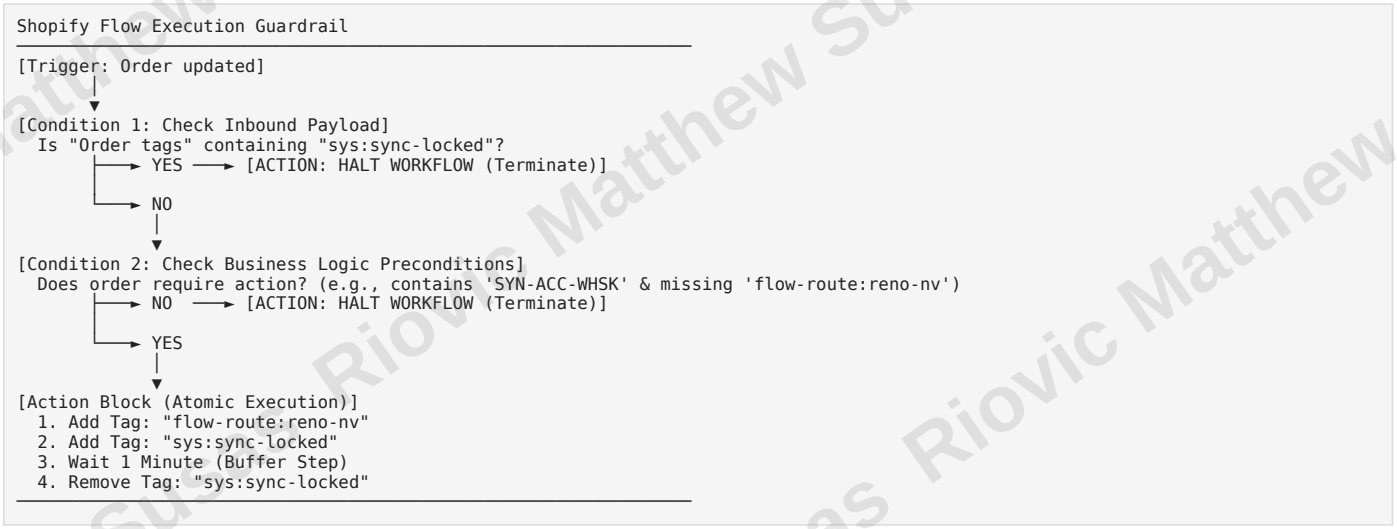
6.1 The Canonical Tag Namespace Standard

All automated processes must adhere to strict reserved tag prefixes. Unprefixed, free-text tags are prohibited from programmatic workflow logic.

Namespace Prefix	Governance Scope	Permitted Actors	Purpose & Syntax Standard	Example Canonical Tags
`flow-hold:`	Order Lifecycle Holds	Shopify Flow (P0), Risk Engines	Temporarily suspends fulfillment. Prevents 3PL transmission.	`flow-hold:fraud-review` `flow-hold:address-invalid` `flow-hold:customer-edit`
`flow-route:`	Physical Node Routing	Shopify Flow (P1), Routing Rules	Directs order allocation to specific fulfillment centers.	`flow-route:allentown-pa` `flow-route:reno-nv` `flow-route:heavy-pack`
`sys:`	Architecture & State Flags	Middleware, Integration Scripts	System state tracking, lock acquisition, idempotency sentinels.	`sys:webhook-processed` `sys:idempotency-locked` `sys:legacy-migrated`
`cs:`	Support Escalations	Gorgias CRM, Support Agents	Customer care routing and customer tier classifications.	`cs:vip-tier-1` `cs:damaged-claim` `cs:manual-override`
`wh:`	Physical Warehouse Ops	3PL WMS, Handheld Scanners	Feedback loop from warehouse floor (packing, waves, tracking).	`wh:wave-assigned` `wh:picked` `wh:box-size-m3`

6.2 Mutual Exclusion & Recursion Guardrail Logic

To prevent an `Order updated` trigger from executing infinitely, every Shopify Flow workflow modifying an order **must implement a dual-stage Mutual Exclusion Check**:



The Four Mandatory Recursion Rules

1. **Never trigger on `Order updated` to execute unconditional tag additions.** The workflow must always verify that the target tag does *not* already exist prior to executing the add action.
2. **Explicit Negative Preconditions:** If Workflow A adds `flow-route:reno-nv`, the very first condition check must be:
   `Order tags` → `does NOT contain` → `flow-route:reno-nv`.
3. **Cross-Workflow Exclusion:** If Workflow A sets `flow-hold:fraud-review` and Workflow B processes fulfillments, Workflow B must include:
   `Order tags` → `does NOT contain` → `flow-hold:*`.
4. **Decoupling via Asynchronous Buffer:** High-frequency tag updates must introduce a `Wait 1 minute` block before stripping intermediate lock tags (`sys:sync-locked`), preventing racing webhooks from executing against dirty cache states.

7. Master Interconnectivity & System Architecture Lineage

The technical matrices in this document do not exist in isolation; they govern the real-time operational state transitions established across the complete Synapse Botanicals TechOps deliverable suite:

Upstream / Downstream Deliverable	Governing Interface / Node	Interconnectivity Mechanism
SOP - OrderExceptionMutationProtocol - Parent (Domain 1.1)	Customer Address Edits & Line Item Swaps	P0 Flow Holds (`flow-hold:customer-edit`) suspend 3PL webhook dispatch while an order edit is processed, preventing dual-fulfillment race conditions.
Matrix - OrderStatePointOfNoReturn - Master (Domain 1.2)	Order State Machine & Fulfillment Gates	When warehouse status transitions to `PROCESSING` (PONR), webhook handlers reject inbound edit payloads with `422 Unprocessable Entity` and tag `sys:edit-rejected-ponr` .
Matrix - OrderRoutingCostOptimization - DecisionEngine (Domain 2.2)	Dynamic Order Allocation Rules	P1 Routing Workflows evaluate SKU availability across Allentown (`LOC-PA-01`) and Reno (`LOC-NV-02`) before emitting fulfillment webhooks to regional WMS nodes.
SOP - BusinessLogicAutomationAndWebhooks - Parent (Domain 4.1)	Architectural Specifications & HMAC Verification	Establishes the foundational standard operating procedures, timing-safe cryptographic security, and emergency kill-switch protocols operationalized by this matrix.
Audit - WebhookFailureAndFlowRecursion - RunSheet (Domain 4.3)	Empirical Simulation & Forensic Run Sheet	Provides the real-world operational testing ground, step-by-step diagnostic audit log, and payload recovery run sheet validating the circuit-breaker and triage rules codified in this engine.